PROJECT-BASED LEARNING

PHASE-I EVALUATION

Cyber Crime Case Management

Team ID: DSCPP-III-2026-T093

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun

Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T093
Semester: 3rd
Section: CORE-F/ML-1/2/4
Domain: AI/ML

Team Members

1. Kusum Sapra – 2510011573
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Mentor

Ms. Ayushi Saini **Interactions:** 2

Problem Statement & Motivation

Problem Statement

Cybercrime investigations involve large amounts of **case, evidence, suspect, officer, and status data**. Manual handling makes information difficult to **search, update, organize, and prioritize**.

Motivation

- Reduce manual record keeping.
- Avoid scattered information.
- Enable faster case retrieval.
- Improve case prioritization.

Target Users

- Cybercrime investigators.
- Police officers and case managers.
- Supervisors and forensic teams.
- Administrative staff.

Goal: Centralize case information for faster search, tracking, and prioritization.

Objectives

Project Objectives

1. Centralize cybercrime case records.
2. Enable fast case search and filtering.
3. Manage evidence and suspects.
4. Track investigation progress.
5. Manage officer assignment.
6. Classify cases by priority.
7. Apply PBL course concepts.
8. Provide future AI/ML scope.

Key Objective

Efficient case management, tracking, and prioritization.

Proposed Solution

Proposed Approach

1. Case Registration
2. Data Processing
3. Case Management
4. Database Storage
5. Reports / Dashboard

Key Features

- Case Registration
- Case Search & Filtering
- Evidence Management
- Officer Assignment
- Status Tracking
- Priority Classification

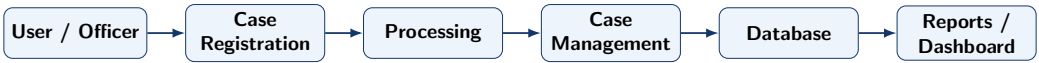
Centralized system for storing, searching, tracking, and prioritizing cybercrime cases.

Integration of PBL Course Concepts

Course	Concepts Applied	Module
Data Structures in C	Arrays, Linked Lists, Hashing, Searching and Sorting	Case Search
OOPs with C++	Classes,Objects,Encpsulation,Inheritance,STL	System Modules

Integration: Data Structures + OOPs

System Architecture / Workflow



Major Modules

- User Management
- Case Management
- Evidence Management
- Officer Management
- Priority Module

Workflow

1. Enter case data
2. Validate and process
3. Store in database
4. Search / update
5. Generate reports

Technology Stack & Methodology

Technology Stack

Programming:	C / C++
Database:	MySQL
Data Structures:	Lists, Hashing, Trees
IDE:	VS Code / Code::Blocks
Version Control:	Git / GitHub
Future AI/ML:	Python / ML

Development Methodology

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Evaluation

Technology Choice: C/C++ for PBL concepts, MySQL for storage, and Git for collaboration.

Initial Progress & Team Contribution

Progress Achieved

- Problem identified
- Requirements analyzed
- Architecture designed
- Modules identified
- Database planned
- PBL concepts mapped

Team Contribution

Member	Role	Contribution
Kusum	Lead	Requirements & design
Ananya	Developer	OOP & case module
Rudransh	DB/Testing	Database & testing
Aryan	Developer	Data structures & docs

Roadmap, Expected Outcomes & References

Roadmap

1. **Phase-I:** Design
2. **Phase-II:** Development
3. **Phase-III:** Testing & Finalization

Expected Outcomes

- Working case management system
- Faster case retrieval
- Organized evidence records
- Case prioritization
- Future AI/ML integration

References

1.Data Structures and Algorithms Made Easy — Narasimha Karumanchi
2.Data Structures Using C — Reema Thareja
3.Object-Oriented Programming with C++ — E. Balagurusamy

Thank You

Questions & Suggestions